KINETICS OF SYSTEM OF PARTICLE

LECTURE: 29-30

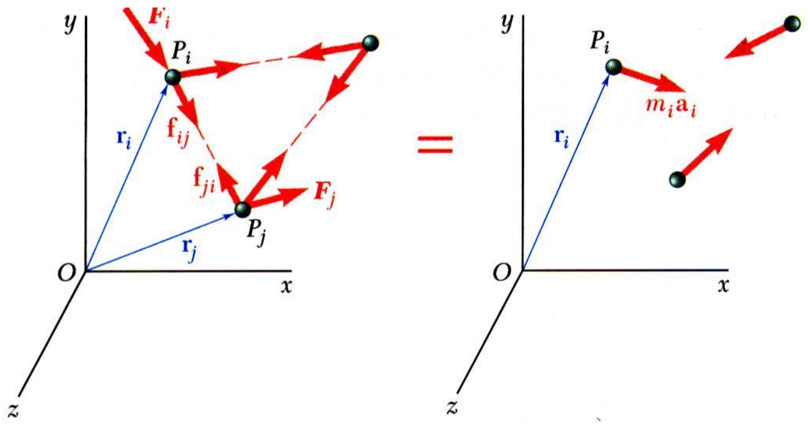
Contents

Engineers often need to analyze the dynamics of systems of particles – this is the basis for many fluid dynamics applications, and will also help to establish the principles used in analyzing rigid bodies

Introduction

- In the current chapter, you will study the motion of *systems of particles*.
- The *effective force* of a particle is defined as the product of it mass and acceleration. It will be shown that the *system of external forces* acting on a system of particles is *equipollent* with the *system of effective forces* of the system.
- The *mass center* of a system of particles will be defined and its motion will be described.
- Application of the *work-energy principle* and the *impulse-momentum principle* to a system of particles will be described. Result obtained are also applicable to a system of rigidly connected particles, i.e., a *rigid body*.
- Analysis methods will be presented for *variable systems of particles*, i.e., systems of the particles change.

Application of Newton's Laws. Effective Forces



- Newton's second law for each particle P_i in a system of n particles,

$$\vec{F}_i + \sum_{j=1}^{n} \vec{f}_{ij} = m_i \vec{a}_i$$

$$\vec{r}_i \times \vec{F}_i + \sum_{j=1}^{n} \left(\vec{r}_i \times \vec{f}_{ij} \right) = \vec{r}_i \times m_i \vec{a}_i$$

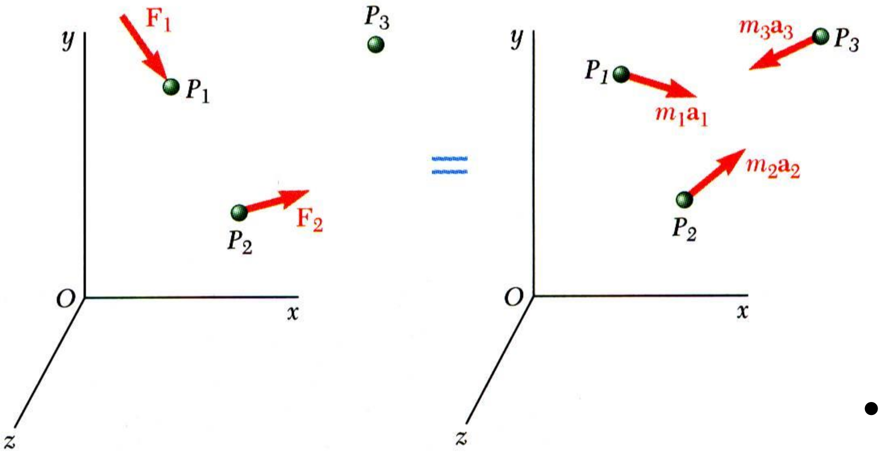
$\vec{F}_i = \text{external force}$ $\qquad$ $\vec{f}_{ij} = \text{internal forces}$

$m_i \vec{a}_i = \text{effective force}$

- The system of external and internal forces on a particle is *equivalent* to the effective force of the particle.

- The system of external and internal forces acting on the entire system of particles is *equivalent* to the system of effective forces.

Application of Newton's Laws. Effective Forces



- Summing over all the elements,

$$\sum_{i=1}^{n} \vec{F}_i + \sum_{i=1}^{n} \sum_{j=1}^{n} \vec{f}_{ij} = \sum_{i=1}^{n} m_i \vec{a}_i$$

$$\sum_{i=1}^{n} \left(\vec{r}_i \times \vec{F}_i\right) + \sum_{i=1}^{n} \sum_{j=1}^{n} \left(\vec{r}_i \times \vec{f}_{ij}\right) = \sum_{i=1}^{n} \left(\vec{r}_i \times m_i \vec{a}_i\right)$$

- Since the internal forces occur in equal and opposite collinear pairs, the resultant force and couple due to the internal forces are zero,

$$\sum \vec{F}_i = \sum m_i \vec{a}_i$$

$$\sum \left(\vec{r}_i \times \vec{F}_i\right) = \sum \left(\vec{r}_i \times m_i \vec{a}_i\right)$$

Linear & Angular Momentum

- Linear momentum of the system of particles,

$$\vec{L} = \sum_{i=1}^{n} m_i \vec{v}_i$$

$$\dot{\vec{L}} = \sum_{i=1}^{n} m_i \dot{\vec{v}}_i = \sum_{i=1}^{n} m_i \vec{a}_i$$

- Resultant of the external forces is equal to rate of change of linear momentum of the system of particles,

$$\sum \vec{F} = \dot{\vec{L}}$$

- Angular momentum about fixed point O of system of particles,

$$\vec{H}_O = \sum_{i=1}^{n} \left(\vec{r}_i \times m_i \vec{v}_i \right)$$

$$\dot{\vec{H}}_O = \sum_{i=1}^{n} \left(\dot{\vec{r}}_i \times m_i \vec{v}_i \right) + \sum_{i=1}^{n} \left(\vec{r}_i \times m_i \dot{\vec{v}}_i \right)$$

$$= \sum_{i=1}^{n} \left(\vec{r}_i \times m_i \vec{a}_i \right)$$

- Moment resultant about fixed point O of the external forces is equal to the rate of change of angular momentum of the system of particles,

$$\sum \vec{M}_O = \dot{\vec{H}}_O$$

Motion of the Mass Center of a System of Particles

- Mass center G of system of particles is defined by position vector $\vec{r}_G$ which satisfies

$$m\vec{r}_G = \sum_{i=1}^{n} m_i \vec{r}_i$$

- Differentiating twice,

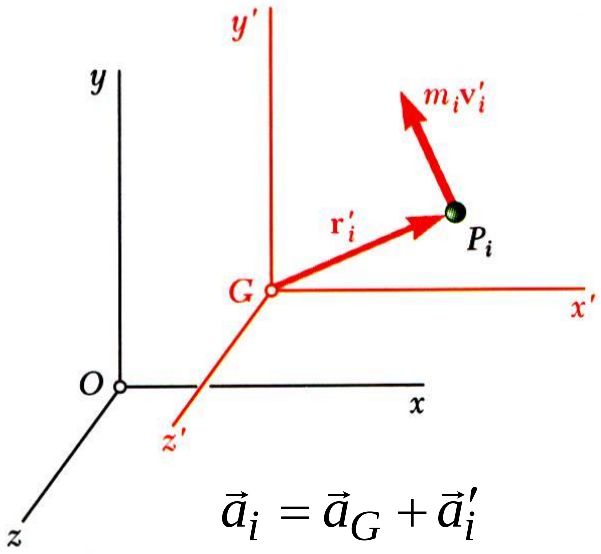
$$m\dot{\vec{r}}_G = \sum_{i=1}^{n} m_i \dot{\vec{r}}_i$$

$$m\vec{v}_G = \sum_{i=1}^{n} m_i \vec{v}_i = \vec{L}$$

$$m\vec{a}_G = \dot{\vec{L}} = \sum \vec{F}$$

- The mass center moves as if the entire mass and all of the external forces were concentrated at that point.

Angular Momentum About the Mass Center



$$\vec{a}_i = \vec{a}_G + \vec{a}'_i$$

- Consider the centroidal frame of reference *Gx'y'z'*, which translates with respect to the Newtonian frame *Oxyz.*

- The centroidal frame is not, in general, a Newtonian frame.

- The angular momentum of the system of particles about the mass center,

$$\vec{H}'_G = \sum_{i=1}^{n}\left(\vec{r}'_i \times m_i \vec{v}'_i\right)$$

$$\dot{\vec{H}}'_G = \sum_{i=1}^{n}\left(\vec{r}'_i \times m_i \vec{a}'_i\right) = \sum_{i=1}^{n}\left(\vec{r}'_i \times m_i\left(\vec{a}_i - \vec{a}_G\right)\right)$$

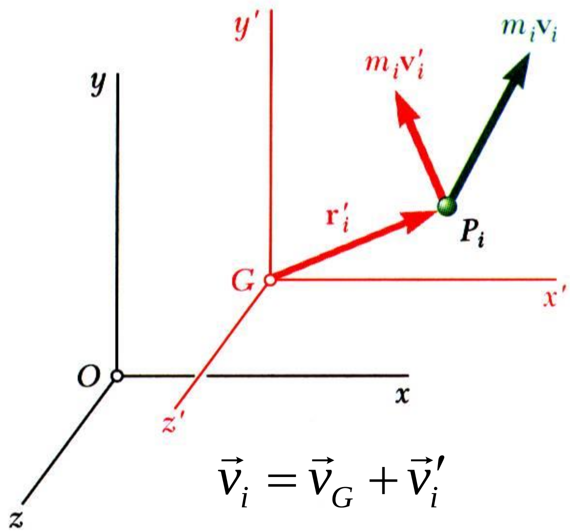
$$= \sum_{i=1}^{n}\left(\vec{r}'_i \times m_i \vec{a}_i\right) - \left(\sum_{i=1}^{n} m_i \vec{r}'_i\right) \times \vec{a}_G$$

$$= \sum_{i=1}^{n}\left(\vec{r}'_i \times m_i \vec{a}_i\right) = \sum_{i=1}^{n}\left(\vec{r}'_i \times \vec{F}_i\right)$$

$$= \sum \vec{M}_G$$

- The moment resultant about *G* of the external forces is equal to the rate of change of angular momentum about *G* of the system of particles.

Angular Momentum About the Mass Center



$$\vec{v}_i = \vec{v}_G + \vec{v}'_i$$

- Angular momentum about G of the particles in their motion relative to the centroidal $Gx'y'z'$ frame of reference,

$$\vec{H}'_G = \sum_{i=1}^{n} \left(\vec{r}'_i \times m_i \vec{v}'_i\right)$$

- Angular momentum about G of particles in their absolute motion relative to the Newtonian $Oxyz$ frame of reference.

$$\begin{aligned}\vec{H}_G &= \sum_{i=1}^{n} \left(\vec{r}'_i \times m_i \vec{v}_i\right) \\ &= \sum_{i=1}^{n} \left(\vec{r}'_i \times m_i \left(\vec{v}_G + \vec{v}'_i\right)\right) \\ &= \left(\sum_{i=1}^{n} m_i \vec{r}'_i\right) \times \vec{v}_G + \sum_{i=1}^{n} \left(\vec{r}'_i \times m_i \vec{v}'_i\right)\end{aligned}$$

$$\vec{H}_G = \vec{H}'_G = \sum \vec{M}_G$$

- Angular momentum about G of the particle can be calculated with respect to either the Newtonian or centroidal frames of reference.

Conservation of Momentum

- If no external forces act on the particles of a system, then the linear momentum and angular momentum about the fixed point *O* are conserved.

$$\dot{\vec{L}} = \sum \vec{F} = 0 \qquad \dot{\vec{H}}_O = \sum \vec{M}_O = 0$$

$$\vec{L} = \text{constant} \qquad \vec{H}_O = \text{constant}$$

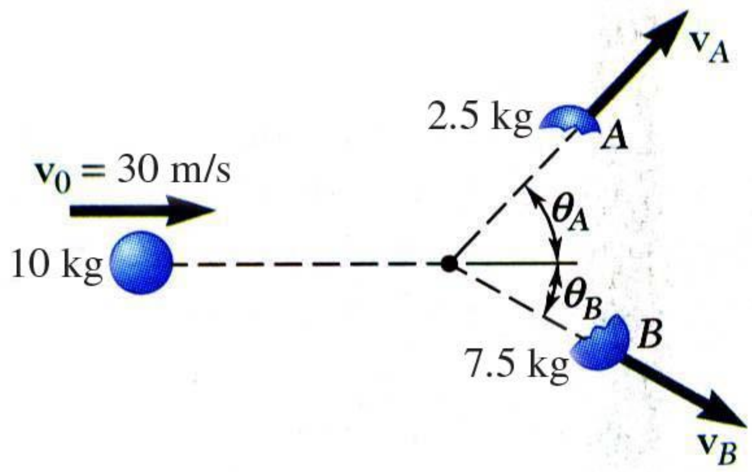
- Concept of conservation of momentum also applies to the analysis of the mass center motion,

$$\dot{\vec{L}} = \sum \vec{F} = 0 \qquad \dot{\vec{H}}_G = \sum \vec{M}_G = 0$$

$$\vec{L} = m\vec{v}_G = \text{constant}$$

$$\vec{v}_G = \text{constant} \qquad \vec{H}_G = \text{constant}$$

Sample Problem 1



A 10-kg projectile is moving with a velocity of 30 m/s when it explodes into 2.5 and 7.5-kg fragments. Immediately after the explosion, the fragments travel in the directions $\theta_A = 45^o$ and $\theta_B = 30^o$.

Determine the velocity of each fragment.

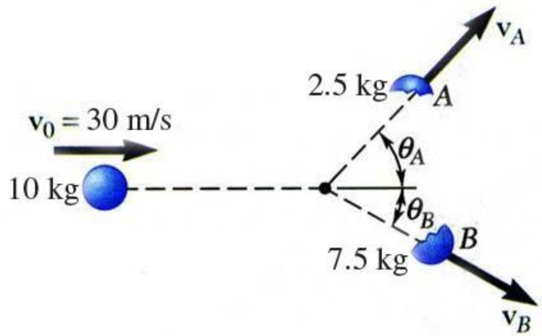
SOLUTION:

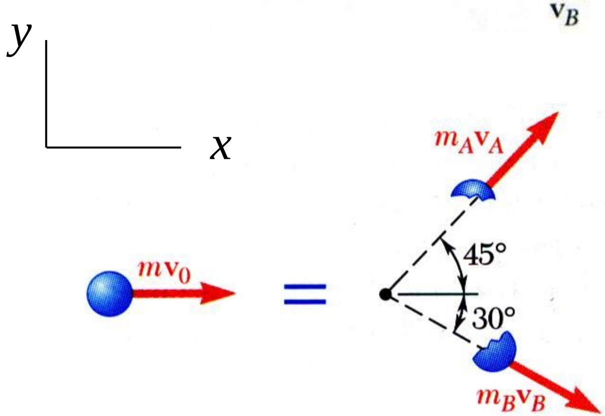
- Since there are no external forces, the linear momentum of the system is conserved.
- Write separate component equations for the conservation of linear momentum.
- Solve the equations simultaneously for the fragment velocities.

Sample Problem 1

SOLUTION:

- Since there are no external forces, the linear momentum of the system is conserved.





- Write separate component equations for the conservation of linear momentum.

$$m_A \vec{v}_A + m_B \vec{v}_B = m\vec{v}_0$$
$$(2.5)\vec{v}_A + (7.5)\vec{v}_B = (10)\vec{v}_0$$

x components:

$$2.5 v_A \cos 45° + 7.5 v_B \cos 30° = 10(30)$$

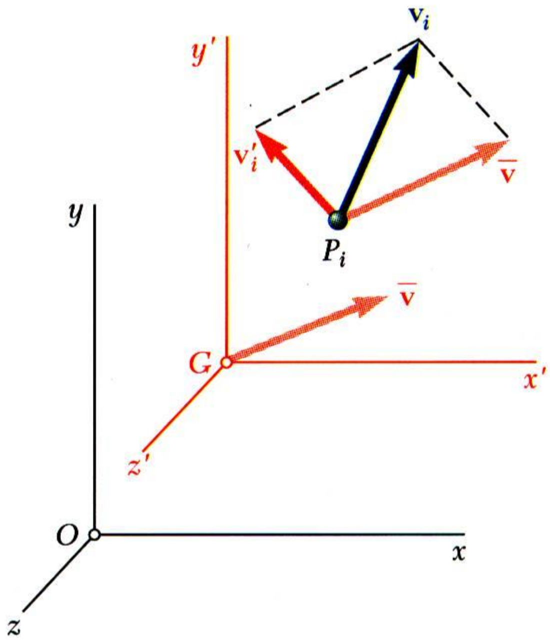
y components:

$$2.5 v_A \sin 45° - 7.5 v_B \sin 30° = 0$$

- Solve the equations simultaneously for the fragment velocities.

$$\boxed{v_A = 62.2\,\mathrm{m/s} \qquad v_B = 29.3\,\mathrm{m/s}}$$

Kinetic Energy



$$\vec{v}_i = \vec{v}_G + \vec{v}'_i$$

- Kinetic energy of a system of particles,

$$T = \frac{1}{2}\sum_{i=1}^{n} m_i(\vec{v}_i \bullet \vec{v}_i) = \frac{1}{2}\sum_{i=1}^{n} m_i v_i^2$$

- Expressing the velocity in terms of the centroidal reference frame,

$$\begin{aligned} T &= \frac{1}{2}\sum_{i=1}^{n}\left[m_i(\vec{v}_G + \vec{v}'_i)\bullet(\vec{v}_G + \vec{v}'_i)\right] \\ &= \frac{1}{2}\left(\sum_{i=1}^{n} m_i\right)v_G^2 + \vec{v}_G \bullet \sum_{i=1}^{n} m_i \vec{v}'_i + \frac{1}{2}\sum_{i=1}^{n} m_i v_i'^2 \\ &= \frac{1}{2} m \vec{v}_G^2 + \frac{1}{2}\sum_{i=1}^{n} m_i v_i'^2 \end{aligned}$$

- Kinetic energy is equal to kinetic energy of mass center plus kinetic energy relative to the centroidal frame.

Work-Energy Principle. Conservation of Energy

- Principle of work and energy can be applied to each particle P_i ,

$$T_1 + U_{1\to 2} = T_2$$

where $U_{1\to 2}$ represents the work done by the internal forces $\vec{f}_{ij}$ and the resultant external force $\vec{F}_i$ acting on P_i .

- Principle of work and energy can be applied to the entire system by adding the kinetic energies of all particles and considering the work done by all external and internal forces.

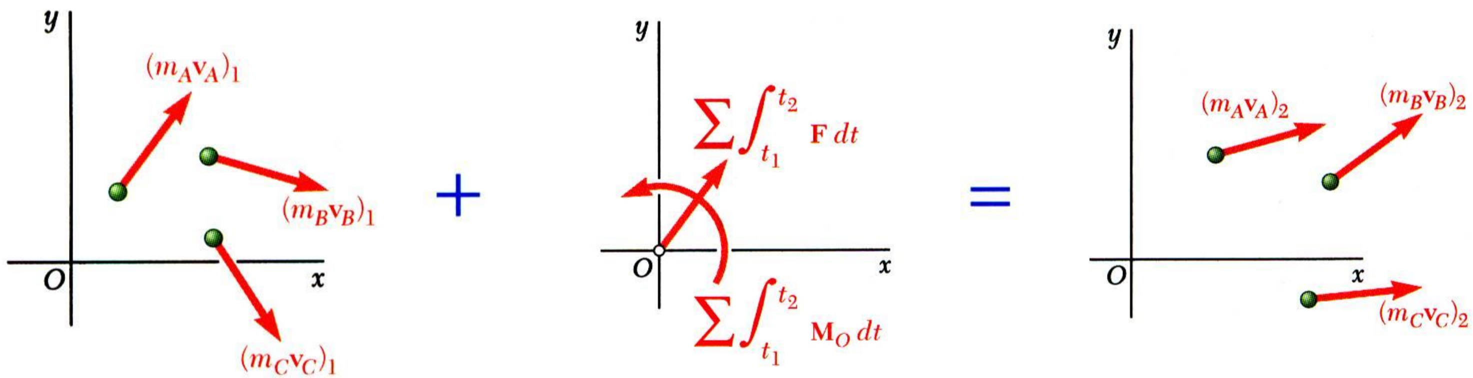
- Although $\vec{f}_{ij}$ and $\vec{f}_{ji}$ are equal and opposite, the work of these forces will not, in general, cancel out.

- If the forces acting on the particles are conservative, the work is equal to the change in potential energy and

$$T_1 + V_1 = T_2 + V_2$$

which expresses the principle of conservation of energy for the system of particles.

Principle of Impulse and Momentum



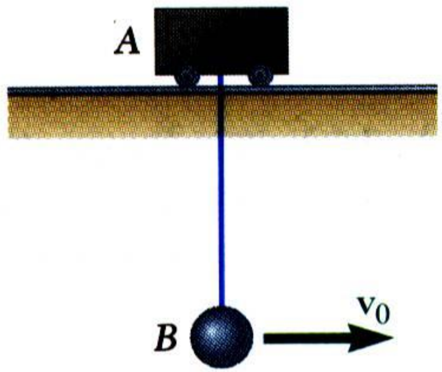
$$\sum \vec{F} = \dot{\vec{L}} \qquad \sum \vec{M}_O = \dot{\vec{H}}_O$$

$$\sum \int_{t_1}^{t_2} \vec{F} dt = \vec{L}_2 - \vec{L}_1 \qquad \sum \int_{t_1}^{t_2} \vec{M}_O dt = \vec{H}_2 - \vec{H}_1$$

$$\vec{L}_1 + \sum \int_{t_1}^{t_2} \vec{F} dt = \vec{L}_2 \qquad \vec{H}_1 + \sum \int_{t_1}^{t_2} \vec{M}_O dt = \vec{H}_2$$

- The momenta of the particles at time t_1 and the impulse of the forces from t_1 to t_2 form a system of vectors *equipollent* to the system of momenta of the particles at time t_2 .

Sample Problem 2



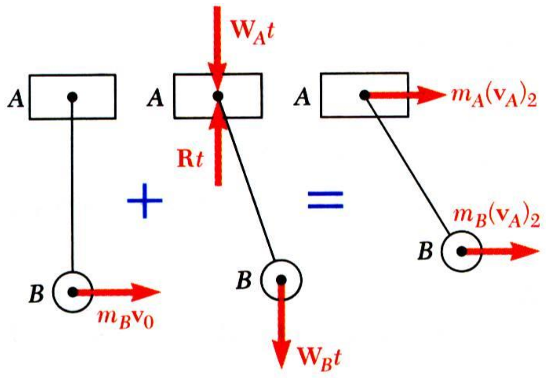
Ball *B*, of mass m_B,is suspended from a chord, of length *l*, attached to cart *A*, of mass m_A, which can roll freely on a frictionless horizontal track. While the cart is at rest, the ball is given an initial velocity $v_0 = \sqrt{2gl}$.

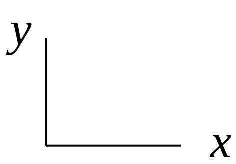
Determine *(a)* the velocity of *B* as it reaches it maximum elevation, and *(b)* the maximum vertical distance *h* through which *B* will rise.

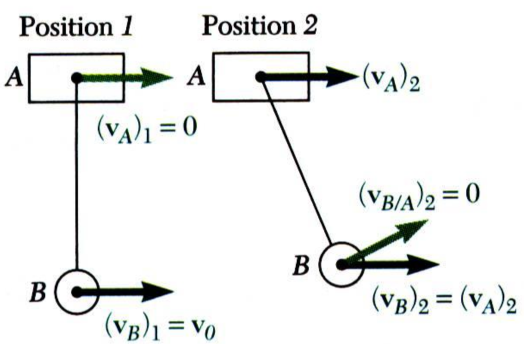
SOLUTION:

- With no external horizontal forces, it follows from the impulse-momentum principle that the horizontal component of momentum is conserved. This relation can be solved for the velocity of *B* at its maximum elevation.
- The conservation of energy principle can be applied to relate the initial kinetic energy to the maximum potential energy. The maximum vertical distance is determined from this relation.

Sample Problem 2







SOLUTION:

- With no external horizontal forces, it follows from the impulse-momentum principle that the horizontal component of momentum is conserved. This relation can be solved for the velocity of *B* at its maximum elevation.

$$\vec{L}_1 + \sum \int_{t_1}^{t_2} \vec{F} dt = \vec{L}_2$$

x component equation:

$$m_A v_{A,1} + m_B v_{B,1} = m_A v_{A,2} + m_B v_{B,2}$$

Velocities at positions 1 and 2 are

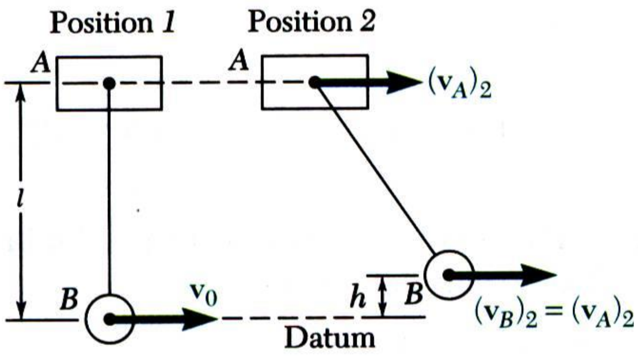
$$v_{A,1} = 0 \quad v_{B,1} = v_0$$

$$v_{B,2} = v_{A,2} + v_{B/A,2} = v_{A,2}$$ (velocity of *B* relative to *A* is zero at position 2)

$$m_B v_0 = (m_A + m_B) v_{A,2}$$

$$\boxed{v_{A,2} = v_{B,2} = \frac{m_B}{m_A + m_B} v_0}$$

Sample Problem 2



- The conservation of energy principle can be applied to relate the initial kinetic energy to the maximum potential energy.

$$T_1 + V_1 = T_2 + V_2$$

Position 1 - Potential Energy: $V_1 = m_A g l$

Kinetic Energy: $T_1 = \frac{1}{2} m_B v_0^2$

Position 2 - Potential Energy: $V_2 = m_A g l + m_B g h$

Kinetic Energy: $T_2 = \frac{1}{2}(m_A + m_B)v_{A,2}^2$

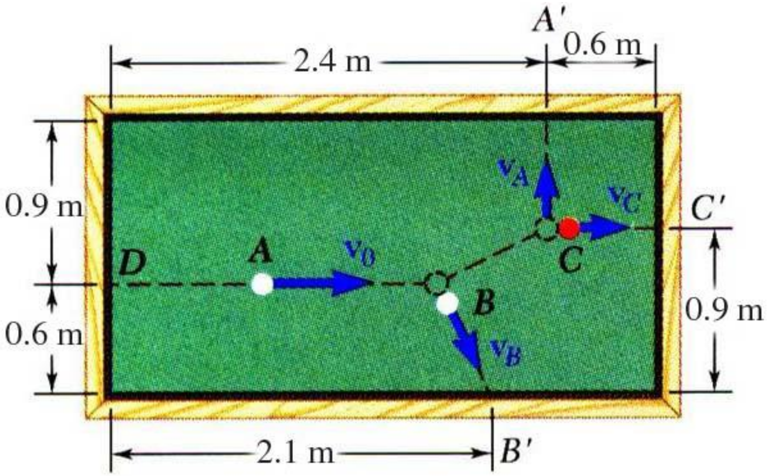
$$\frac{1}{2} m_B v_0^2 + m_A g l = \frac{1}{2}(m_A + m_B)v_{A,2}^2 + m_A g l + m_B g h$$

$$h = \frac{v_0^2}{2g} - \frac{m_A + m_B}{m_B}\frac{v_{A,2}^2}{2g} = \frac{v_0^2}{2g} - \frac{m_A + m_B}{2g\, m_B}\left(\frac{m_B}{m_A + m_B} v_0\right)^2$$

$$h = \frac{v_0^2}{2g} - \frac{m_B}{m_A + m_B}\frac{v_0^2}{2g}$$

$$\boxed{h = \frac{m_A}{m_A + m_B}\frac{v_0^2}{2g}}$$

Sample Problem



Ball *A* has initial velocity v_0 = 3 m/s parallel to the axis of the table. It hits ball *B* and then ball *C* which are both at rest. Balls *A* and *C* hit the sides of the table squarely at *A′* and *C′* and ball *B* hits obliquely at *B′*.

Assuming perfectly elastic collisions, determine velocities v_A, v_B, and v_C with which the balls hit the sides of the table.

SOLUTION:

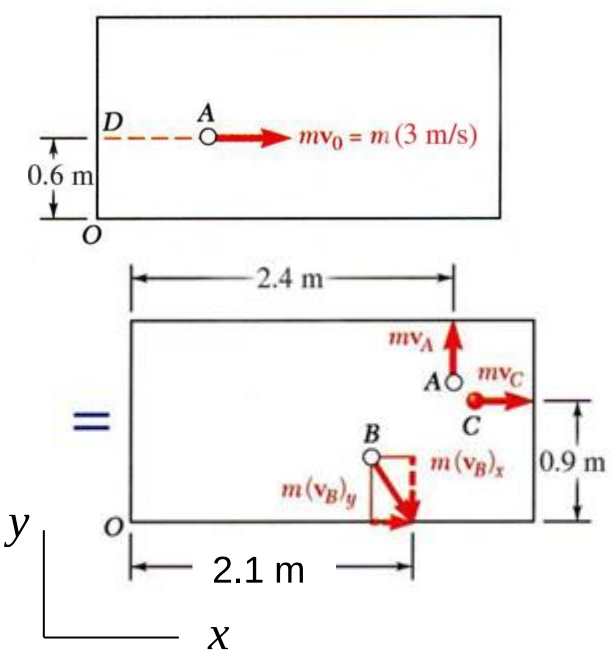
- There are four unknowns: v_A, $v_{B,x}$, $v_{B,y}$, and v_C.
- Solution requires four equations: conservation principles for linear momentum (two component equations), angular momentum, and energy.
- Write the conservation equations in terms of the unknown velocities and solve simultaneously.

Sample Problem

SOLUTION:

- There are four unknowns: v_A, $v_{B,x}$, $v_{B,y}$, and v_C.

$$\vec{v}_A = v_A\vec{j}$$
$$\vec{v}_B = v_{B,x}\vec{i} + v_{B,y}\vec{j}$$
$$\vec{v}_C = v_C\vec{i}$$



- The conservation of momentum and energy equations,

$$\vec{L}_1 + \sum\int \vec{F}dt = \vec{L}_2$$

$$mv_0 = mv_{B,x} + mv_C \qquad 0 = mv_A - mv_{B,y}$$

$$\vec{H}_{O,1} + \sum\int \vec{M}_O\, dt = \vec{H}_{O,2}$$

$$-(0.6\text{m})mv_0 = (2.4\text{m})mv_A - (2.1\text{m})mv_{B,y} - (0.9\text{m})mv_C$$

$$T_1 + V_1 = T_2 + V_2$$

$$\tfrac{1}{2}mv_0^2 = \tfrac{1}{2}mv_A^2 + \tfrac{1}{2}m\left(v_{B,x}^2 + v_{B,y}^2\right) + \tfrac{1}{2}mv_C^2$$

Solving the first three equations in terms of v_C,

$$v_A = v_{B,y} = 3v_C - 6 \qquad v_{B,x} = 3 - v_C$$

Substituting into the energy equation,

$$2\left(3v_C - 6\right)^2 + \left(3 - v_C\right)^2 + v_C^2 = 9$$

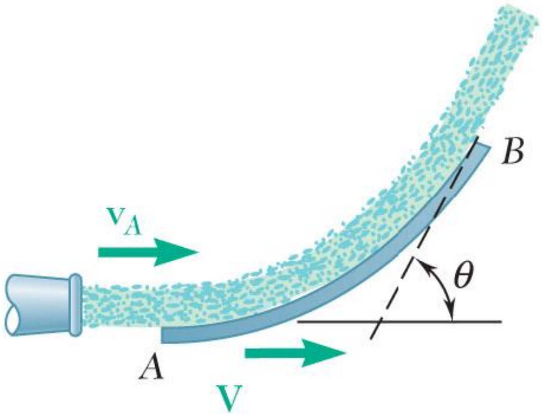
$$20v_C^2 - 78v_C + 72 = 0$$

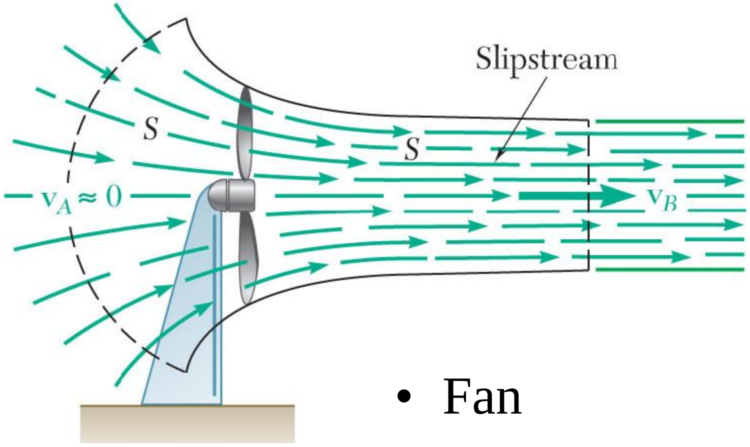
$$\boxed{\begin{aligned} &v_A = 1.2\,\text{m/s} \quad v_C = 2.4\,\text{m/s} \\ &\vec{v}_B = \left(0.6\vec{i} - 1.2\vec{j}\right)\text{m/s} \quad v_B = 1.342\,\text{m/s}\end{aligned}}$$

Variable Systems of Particles

- Kinetics principles established so far were derived for constant systems of particles, i.e., systems which neither gain nor lose particles.
- A large number of engineering applications require the consideration of variable systems of particles, e.g., hydraulic turbine, rocket engine, etc.
- For analyses, consider auxiliary systems which consist of the particles instantaneously within the system plus the particles that enter or leave the system during a short time interval.

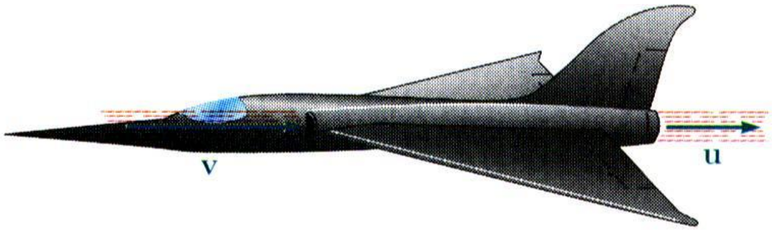
Steady Stream of Particles. Applications



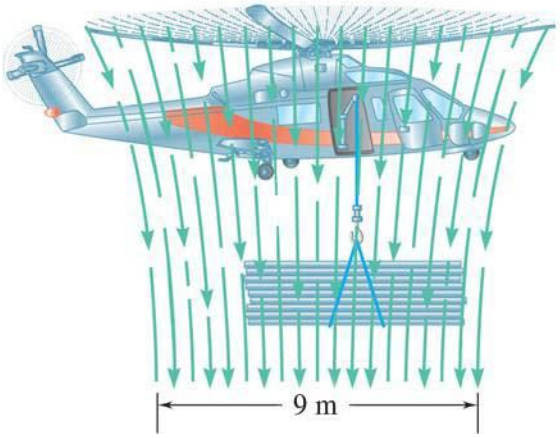


- Fan

- Fluid Stream Diverted by Vane or Duct

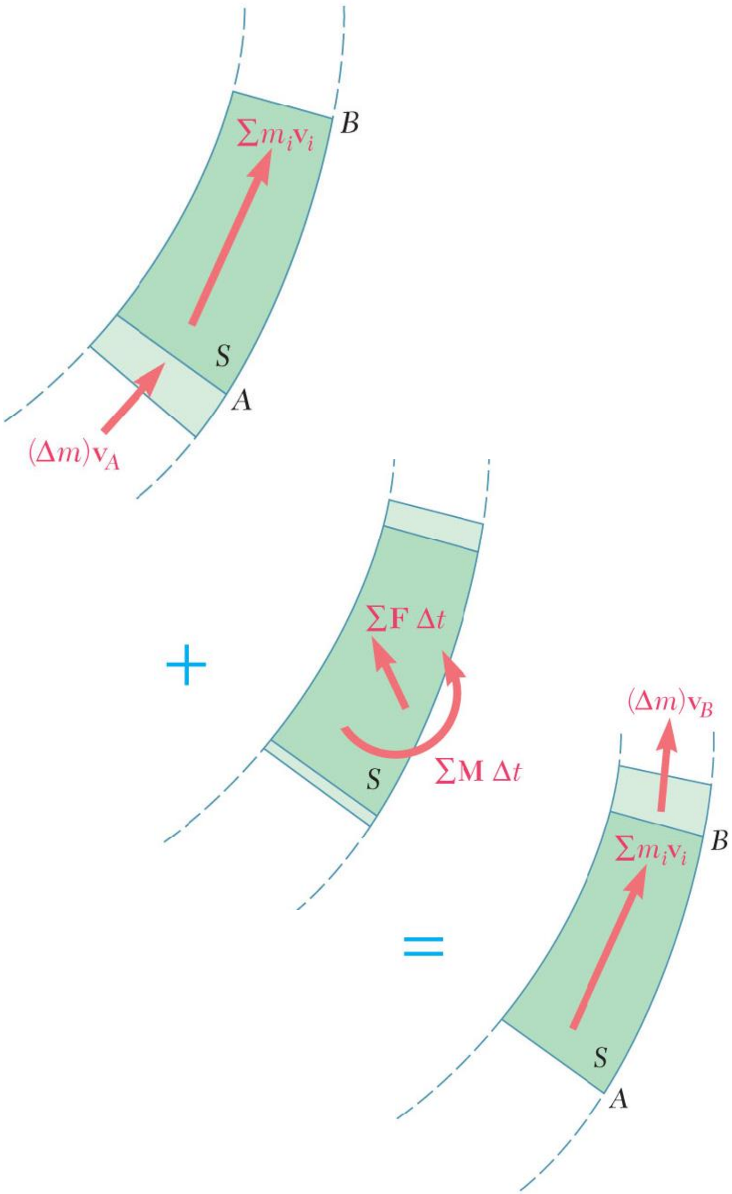


- Jet Engine



- Helicopter

Steady Stream of Particles



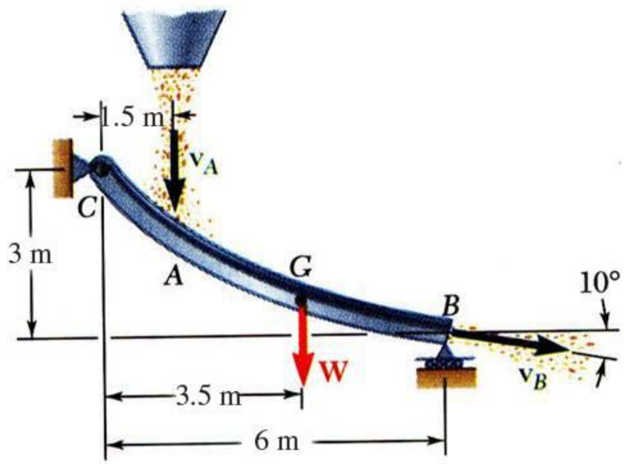
- System consists of a steady stream of particles against a vane or through a duct.
- Define auxiliary system which includes particles which flow in and out over Δt.
- The auxiliary system is a constant system of particles over Δt.

$$\vec{L}_1 + \sum \int_{t_1}^{t_2} \vec{F} dt = \vec{L}_2$$

$$\left[\sum m_i \vec{v}_i + (\Delta m)\vec{v}_A\right] + \sum \vec{F}\,\Delta t = \left[\sum m_i \vec{v}_i + (\Delta m)\vec{v}_B\right]$$

$$\sum \vec{F} = \frac{dm}{dt}\left(\vec{v}_B - \vec{v}_A\right)$$

Sample Problem - 1



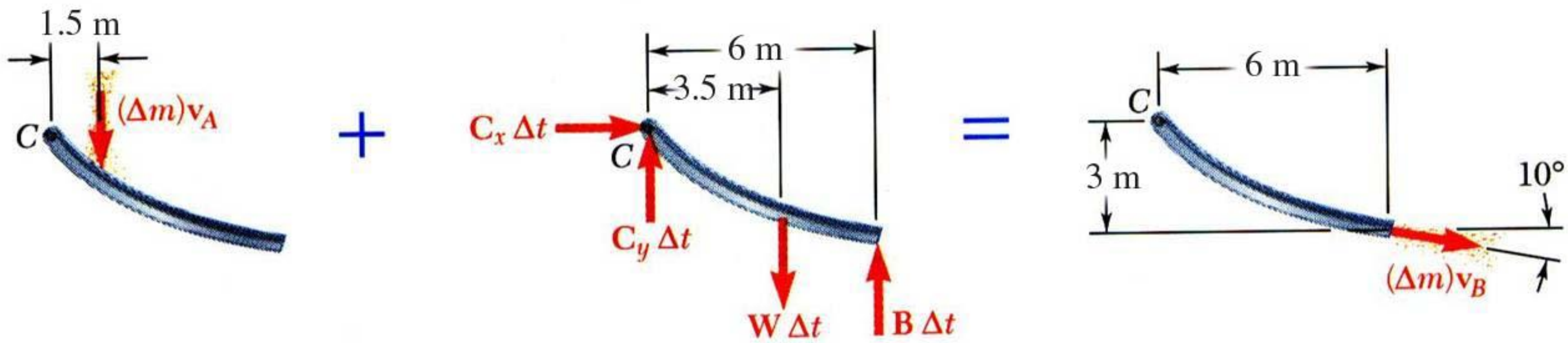
Grain falls onto a chute at the rate of 120 kg/s. It hits the chute with a velocity of 10 m/s and leaves with a velocity of 7.5 m/s. The combined weight of the chute and the grain it carries is 3000 N with the center of gravity at *G*.

Determine the reactions at *C* and *B*.

SOLUTION:

- Define a system consisting of the mass of grain on the chute plus the mass that is added and removed during the time interval Δt.
- Apply the principles of conservation of linear and angular momentum for three equations for the three unknown reactions.

Sample Problem – 1



SOLUTION:

- Define a system consisting of the mass of grain on the chute plus the mass that is added and removed during the time interval Δ*t*.
- Apply the principles of conservation of linear and angular momentum for three equations for the three unknown reactions.

$$\vec{L}_1 + \sum \int \vec{F} dt = \vec{L}_2$$

$$C_x \Delta t = (\Delta m) v_B \cos 10°$$

$$-(\Delta m) v_A + (C_y - W + B)\Delta t = -(\Delta m) v_B \sin 10°$$

$$\vec{H}_{C,1} + \sum \int \vec{M}_C \, dt = \vec{H}_{C,2}$$

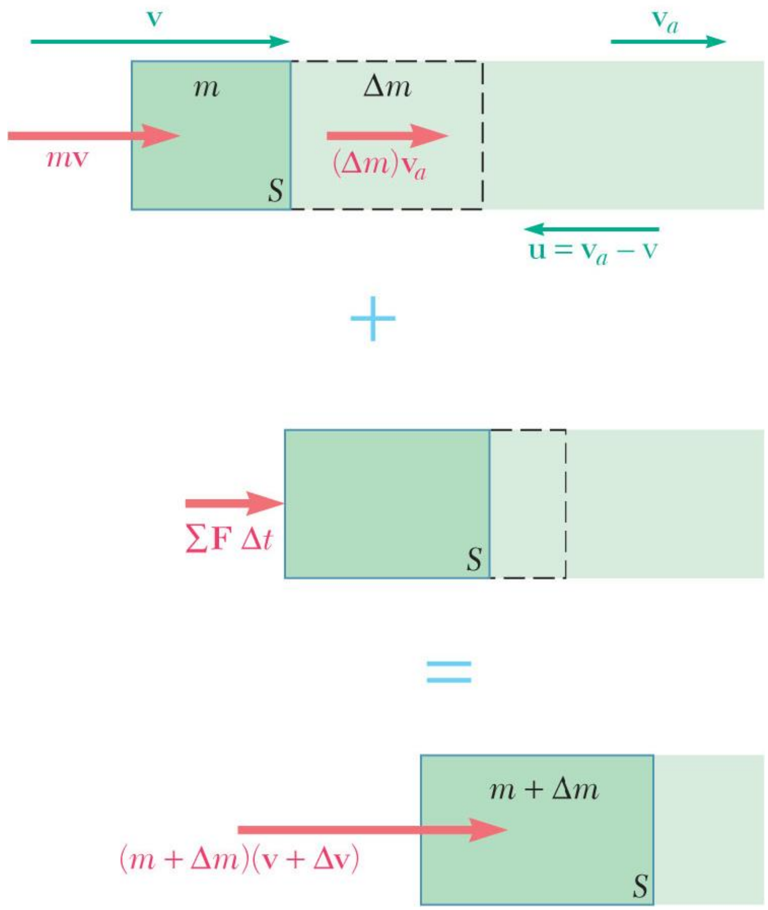
$$-1.5(\Delta m) v_A + (-3.5W + 6B)\Delta t$$
$$= 3(\Delta m) v_B \cos 10° - 6(\Delta m) v_B \sin 10°$$

Solve for C_x, C_y, and B with

$$\frac{\Delta m}{\Delta t} = 120 \text{ kg/s}$$

$$\boxed{\vec{B} = 2336.9\hat{\jmath} \text{ N}, \ \vec{C} = 886.33\hat{\imath} + 1706.8\hat{\jmath} \text{ N}}$$

Streams Gaining or Losing Mass



- Define auxiliary system S to include particles of mass m within system at time t plus the particles of mass Δm which enter the system over time interval Δt.
- The auxiliary system is a constant system of particles.

$$\vec{L}_1 + \sum \int_{t_1}^{t_2} \vec{F} dt = \vec{L}_2$$

$$\left[m\vec{v} + (\Delta m)\vec{v}_a\right] + \sum \vec{F}\,\Delta t = (m + \Delta m)(\vec{v} + \Delta\vec{v})$$

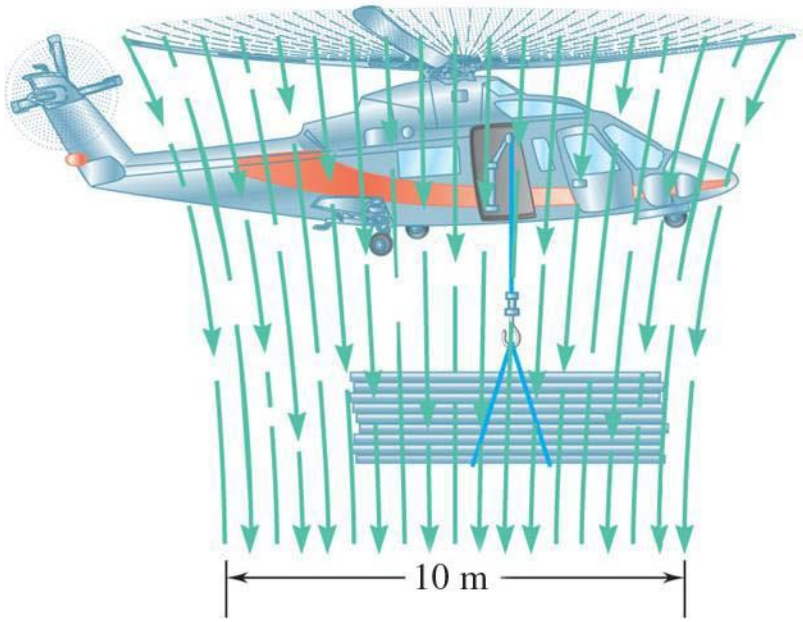
$$\sum \vec{F}\,\Delta t = m\Delta\vec{v} + \Delta m(\vec{v} - \vec{v}_a) + (\Delta m)\Delta\vec{v}$$

Now $\vec{u} = \overrightarrow{v_a} - \vec{v}$ is relative velocity of the absorbed particle with respect to S.

$$\sum \vec{F} = m\frac{d\vec{v}}{dt} - \frac{dm}{dt}\vec{u}$$

$$m\vec{a} = \sum \vec{F} + \frac{dm}{dt}\vec{u}$$

Sample Problem -2



SOLUTION:

- Calculate the time rate of change of the mass of the air.
- Determine the thrust generated by the airstream.
- Use this thrust to determine the maximum load that the helicopter can carry.

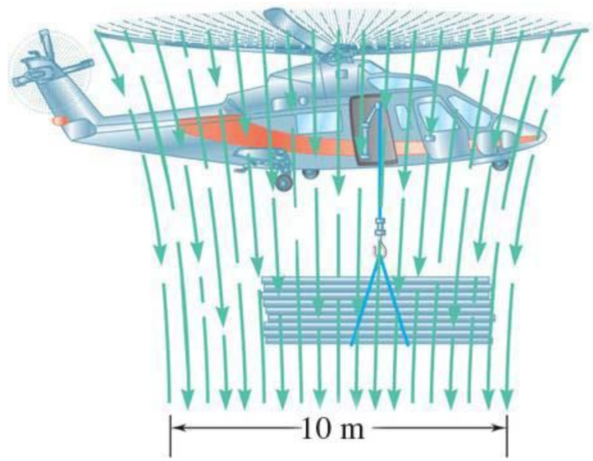
The helicopter shown can produce a maximum downward air speed of *25 m/s* in a 10-m-diameter slipstream. Knowing that the weight of the helicopter and its crew is 18 kN and assuming $\rho = 1.21\ kg/m^3$ for air, determine the maximum load that the helicopter can lift while hovering in midair.

Sample Problem -2

<u>SOLUTION</u>:

Given: v_B = 25 m/s, W= 18000 N, ρ= 1.21 kg/m^3

Find: Max load during hover



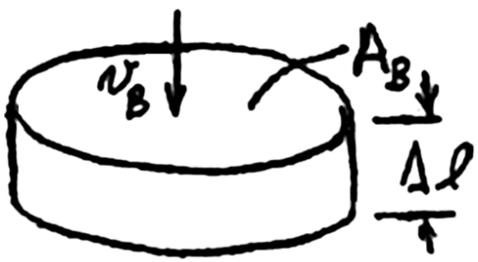
Choose the relationship to determine the thrust

$$F = -\frac{dm}{dt}(v_B - v_A)$$

Calculate the time rate of change (dm/dt) of the mass of the air.

$$\text{mass} = \text{density} \times \text{volume} = \text{density} \times \text{area} \times \text{length}$$

$$\Delta m = \rho A_B(\Delta l) \quad = \rho A_B v_B(\Delta t)$$



$$\frac{dm}{dt} = \rho\, A_B v_B$$

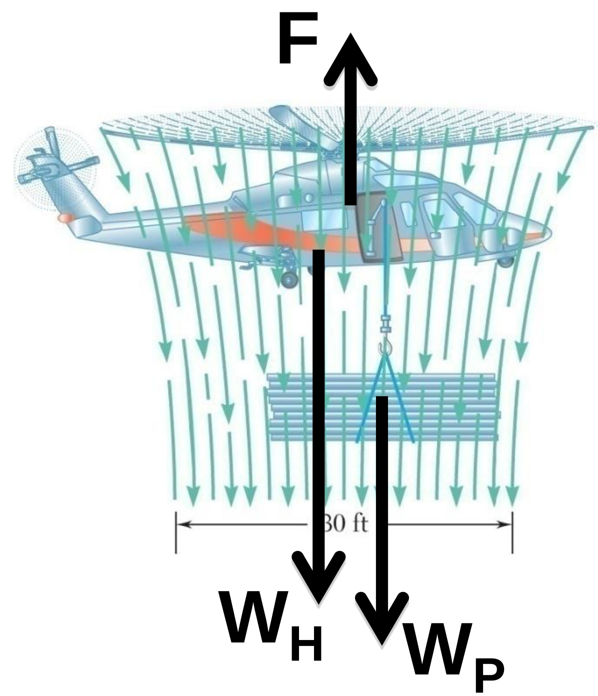
A_B is the area of the slipstream
v_B is the velocity in the slipstream.
Well above the blade, $v_A \approx 0$

Sample Problem -2

Use the relationship for dm/dt to determine the thrust

$$F = -\rho A_B {v_B}^2$$

$$F = -1.21 \times \frac{\pi}{4} 10^2 \times 25^2 = -59396$$



Use statics to determine the maximum payload during hover

$$+\uparrow \sum F_y = F - W_H - W_P = 0$$

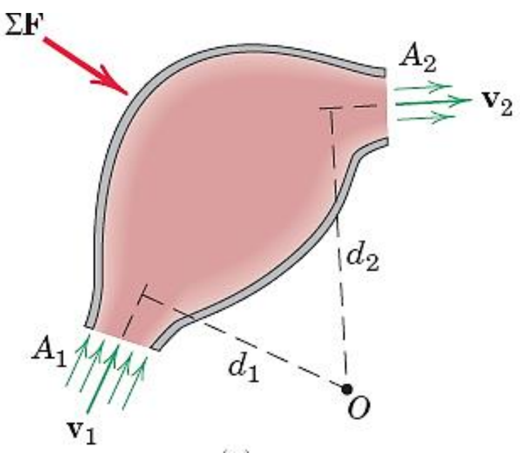
$$W_P = F - W_H = 59{,}396 - 8000 = 41396 \text{ N}$$

$\boxed{W = 41400 \text{ N}}$

Flow Through a Rigid Container

Consider a rigid container, shown in Fig, into which mass flows in a steady stream at the rate ***m'*** through the entrance section of area *A*1. Mass leaves the container through the exit section of area *A*2 at the same rate, so that there is no accumulation or depletion of the total mass within the container during the period of observation.

The velocity of the entering stream is **v**1 normal to *A*1 and that of the leaving stream is **v**2 normal to *A*2. If ρ_1 and ρ_2 are the respective densities of the two streams, conservation of mass requires that



$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2 = m'$$

Flow Through a Rigid Container

The expression for $\dot{\mathbf{G}}$ may be obtained by an incremental analysis.

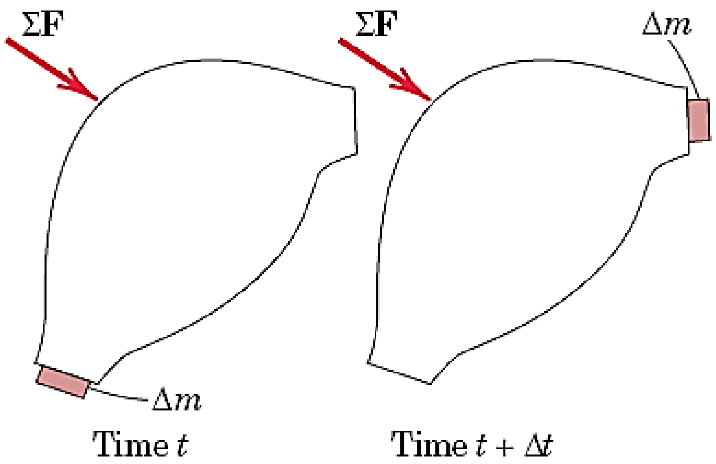


Figure illustrates the system at time t when the system mass is that of the container, the mass within it, and an increment Δm about to enter during time Δt. At time $t + \Delta t$ the same total mass is that of the container, the mass within it, and an equal increment Δm which leaves the container in time Δt.

Incremental Analysis

The linear momentum of the container and mass within it between the two sections A_1 and A_2 remains unchanged during Δt so that the change in momentum of the system in time Δt is

$$\Delta \mathbf{G} = (\Delta m)\mathbf{v}_2 - (\Delta m)\mathbf{v}_1 = \Delta m(\mathbf{v}_2 - \mathbf{v}_1)$$

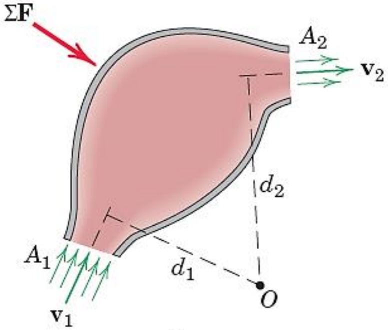
Division by Δt and passage to the limit yield $\dot{\mathbf{G}} = m'\Delta \mathbf{v}$, where

$$m' = \lim_{\Delta t \to 0} \left(\frac{\Delta m}{\Delta t}\right) = \frac{dm}{dt}$$

Thus,

$$\boxed{\Sigma \mathbf{F} = m'\Delta \mathbf{v}}$$

Angular Momentum in Steady-Flow Systems



The resultant moment of all external forces about some fixed point O on or off the system, equals the time rate of change of angular momentum of the system about O, for the case of steady flow in a single plane,

$$\Sigma M_O = m'(v_2 d_2 - v_1 d_1)$$

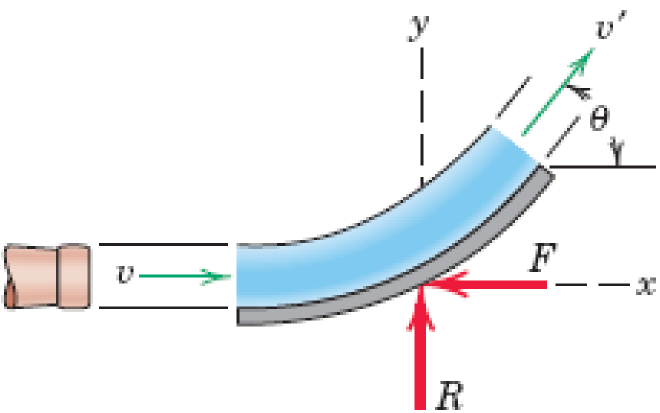
When the velocities of the incoming and outgoing flows are not in the same plane, the equation may be written in vector form as

$$\boxed{\Sigma \mathbf{M}_O = m'(\mathbf{d}_2 \times \mathbf{v}_2 - \mathbf{d}_1 \times \mathbf{v}_1)}$$

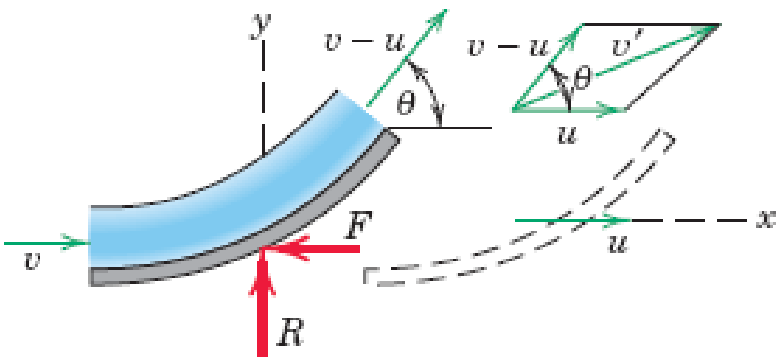
where $\mathbf{d}_1$ and $\mathbf{d}_2$ are the position vectors to the centers of A_1 and A_2 from the fixed reference O. In both relations, the mass center G may be used alternatively as a moment center

Sample Problem -3

The smooth vane shown diverts the open stream of fluid of cross-sectional area A, mass density ρ, and velocity v. (a) Determine the force components R and F required to hold the vane in a fixed position. (b) Find the forces when the vane is given a constant velocity u less than v and in the direction of v.



Fixed vane



Moving vane

Sample Problem -3

Solution. Part (a). The free-body diagram of the vane together with the fluid portion undergoing the momentum change is shown. The momentum equation may be applied to the isolated system for the change in motion in both the x- and y-directions. With the vane stationary, the magnitude of the exit velocity v' equals that of the entering velocity v with fluid friction neglected. The changes in the velocity components are then

$$\Delta v_x = v' \cos\theta - v = -v(1 - \cos\theta)$$

and

$$\Delta v_y = v' \sin\theta - 0 = v \sin\theta$$

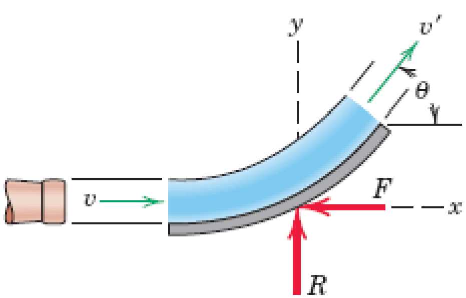
The mass rate of flow is $m' = \rho A v$, and substitution into $\Sigma \mathbf{F} = m' \Delta \mathbf{v}$

$[\Sigma F_x = m' \Delta v_x]$ $\qquad -F = \rho A v[-v(1 - \cos\theta)]$

$$F = \rho A v^2 (1 - \cos\theta)$$

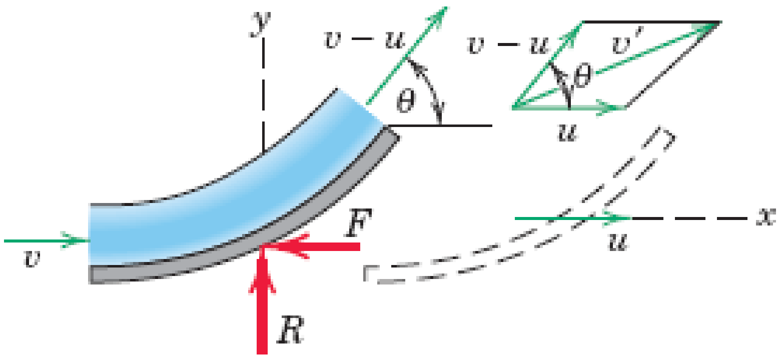
$[\Sigma F_y = m' \Delta v_y]$ $\qquad R = \rho A v[v \sin\theta]$

$$R = \rho A v^2 \sin\theta$$



Fixed vane

Sample Problem -3



Moving vane

Part (b). In the case of the moving vane, the final velocity v' of the fluid upon exit is the vector sum of the velocity u of the vane plus the velocity of the fluid relative to the vane $v - u$. This combination is shown in the velocity diagram to the right of the figure for the exit conditions. The x-component of v' is the sum of the components of its two parts, so $v'_x = (v - u)\cos\theta + u$. The change in x-velocity of the stream is

$$\Delta v_x = (v - u)\cos\theta + (u - v) = -(v - u)(1 - \cos\theta)$$

The y-component of v' is $(v - u)\sin\theta$, so that the change in the y-velocity of the stream is $\Delta v_y = (v - u)\sin\theta$.

Sample Problem -3

The mass rate of flow m' is the mass undergoing momentum change per unit of time. This rate is the mass flowing over the vane per unit time and *not* the rate of issuance from the nozzle. Thus,

$$m' = \rho A(v - u)$$

The impulse-momentum principle of applied in the positive coordi-nate directions gives

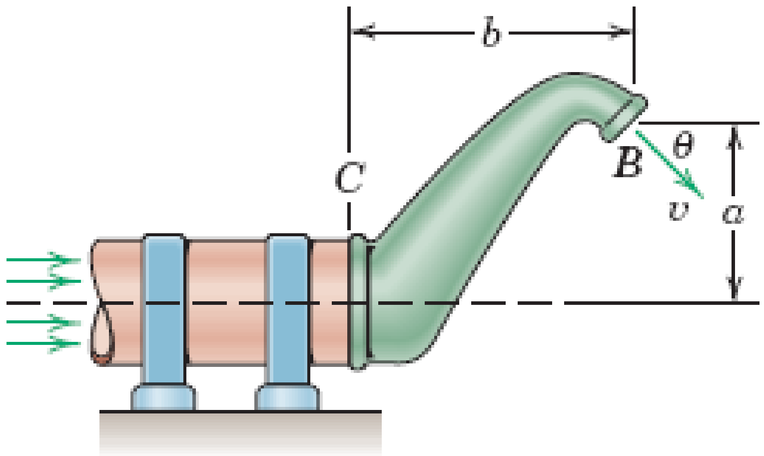
$$[\Sigma F_x = m'\Delta v_x] \qquad -F = \rho A(v - u)[-(v - u)(1 - \cos\theta)]$$

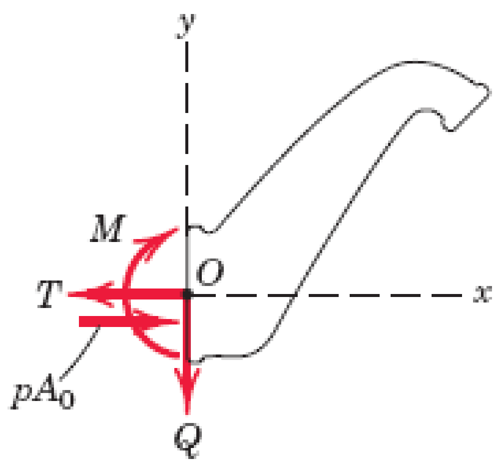
$$F = \rho A(v - u)^2(1 - \cos\theta) \qquad Ans.$$

$$[\Sigma F_y = m'\Delta v_y] \qquad R = \rho A(v - u)^2 \sin\theta \qquad Ans.$$

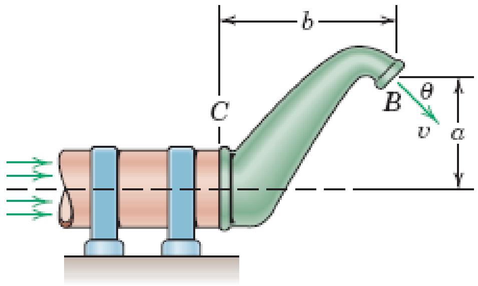
Sample Problem -4

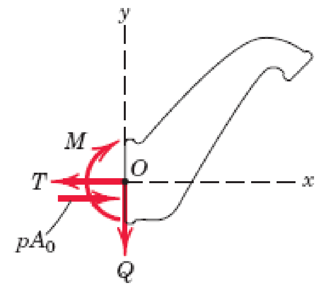
The offset nozzle has a discharge area A at B and an inlet area A_0 at C. A liquid enters the nozzle at a static gage pressure p through the fixed pipe and issues from the nozzle with a velocity v in the direction shown. If the constant density of the liquid is ρ, write expressions for the tension T, shear Q, and bending moment M in the pipe at C.





Sample Problem -4





Solution. The free-body diagram of the nozzle and the fluid within it shows the tension T, shear Q, and bending moment M acting on the flange of the nozzle where it attaches to the fixed pipe. The force pA_0 on the fluid within the nozzle due to the static pressure is an additional external force.

Continuity of flow with constant density requires that

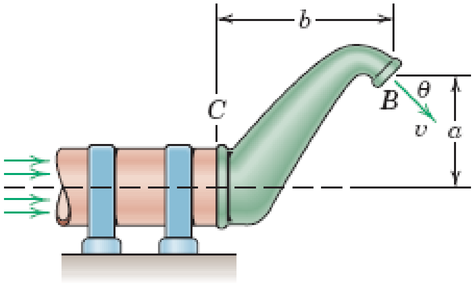
$$Av = A_0 v_0$$

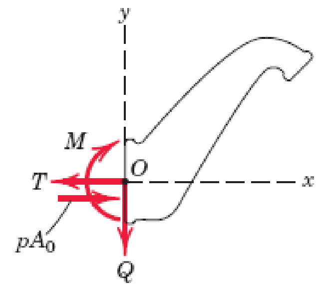
where v_0 is the velocity of the fluid at the entrance to the nozzle. The momentum principle of applied to the system in the two coordinate directions gives

$[\Sigma F_x = m'\Delta v_x]$ $\qquad pA_0 - T = \rho Av(v \cos\theta - v_0)$

$$T = pA_0 + \rho Av^2 \left(\frac{A}{A_0} - \cos\theta \right) \qquad \textit{Ans.}$$

Sample Problem -4





$$T = pA_0 + \rho A v^2 \left(\frac{A}{A_0} - \cos\theta \right)$$

$[\Sigma F_y = m' \Delta v_y]$ $$-Q = \rho A v(-v \sin\theta - 0)$$

$$Q = \rho A v^2 \sin\theta$$

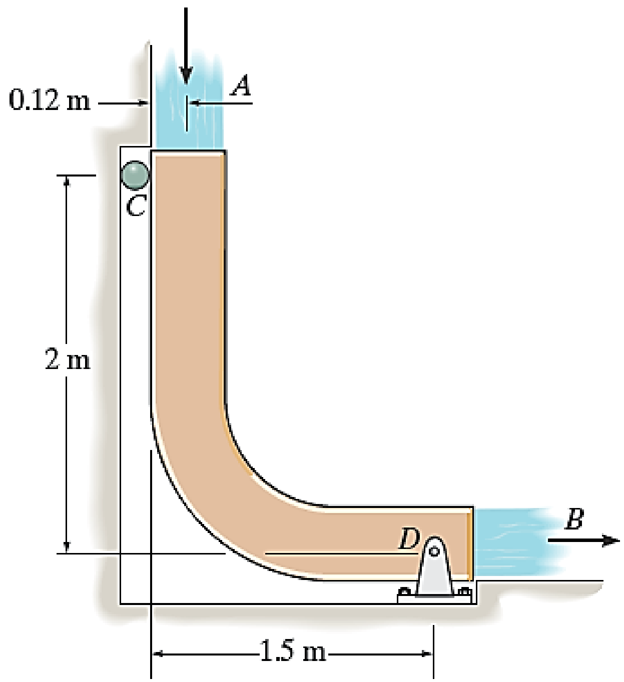
The moment principle of applied in the clockwise sense gives

$[\Sigma M_O = m'(v_2 d_2 - v_1 d_1)]$ $$M = \rho A v(va \cos\theta + vb \sin\theta - 0)$$

$$M = \rho A v^2 (a \cos\theta + b \sin\theta)$$

Steady Mass Flow: Sample Problem -5

The chute is used to divert the flow of water, $Q = 0.6\ \mathrm{m^3/s}$. If the water has a cross-sectional area of $0.05\ \mathrm{m^2}$, determine the force components at the pin D and roller C necessary for equilibrium. Neglect the weight of the chute and weight of the water on the chute, $\rho_w = 1\ \mathrm{Mg/m^3}$.



Steady Mass Flow: Sample Problem -5

Equations of Steady Flow: Here, the flow rate $Q = 0.6\ \text{m}^2/\text{s}$. Then, $v = \dfrac{Q}{A} = \dfrac{0.6}{0.05} = 12.0\ \text{m/s}$. Also, $\dfrac{dm}{dt} = \rho_w Q = 1000\,(0.6) = 600\ \text{kg/s}$.

$$\circlearrowleft + \Sigma M_B = \frac{dm}{dt}(d_{DB}\, v_B - d_{DA}\, v_A);$$

$$-C_x(2) = 600\,[0 - 1.38(12.0)] \qquad C_x = 4968\ \text{N} = 4.97\ \text{kN}$$

$$\xrightarrow{+} \Sigma F_x = \frac{dm}{dt}\left(v_{B_x} - v_{A_x}\right);$$

$$D_x + 4968 = 600\,(12.0 - 0) \qquad D_x = 2232\text{N} = 2.23\ \text{kN}$$

$$+\uparrow \Sigma F_y = \Sigma \frac{dm}{dt}\left(v_{\text{out}_y} - v_{\text{in}_y}\right);$$

$$D_y = 600[0 - (-12.0)] \qquad D_y = 7200\ \text{N} = 7.20\ \text{kN}$$